

CASE STUDY REPORT

EV Battery Degradation, Range Anxiety & User Transparency System

Analysing battery health, driving range and a proposed
transparent monitoring dashboard for EV users

FOCUS AREA

Battery Health & Range Anxiety

BATTERY TYPE

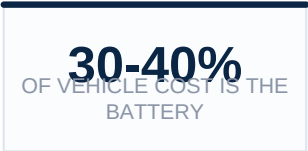
Lithium-Ion (Li-ion)

KEY METRIC

State of Health (SOH)

Table of Contents

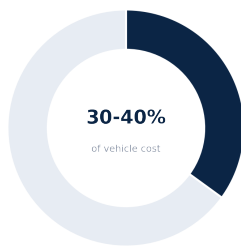
1. Introduction & Objectives
2. Overview of EV Batteries & Current Problem
3. Causes of Battery Degradation
4. Battery Life Cycle Analysis
5. Temperature & Charging Behaviour Effects
6. Range Anxiety Analysis
7. Proposed User Transparency System
8. Comparison, Advantages & Limitations
9. Results & Conclusion



01 Introduction & Objectives

Electric Vehicles (EVs) are becoming an essential part of sustainable transportation. The battery is the most critical component of an EV, accounting for nearly **30–40% of the vehicle's total cost**. Most modern EVs use Lithium-Ion batteries because of their high energy density, long life cycle, and fast charging capability.

As batteries age, their storage capacity gradually decreases — a phenomenon known as **battery degradation** — resulting in reduced driving range, longer charging times, and lower overall vehicle performance. Since most EV dashboards only display battery percentage and estimated range, users have limited information about the actual condition of their battery. This lack of transparency contributes significantly to **range anxiety**.



Aim

To analyze the factors affecting EV battery degradation, understand its impact on driving range and user confidence, and propose a transparent battery monitoring system that provides users with accurate battery health information to reduce range anxiety.

Objectives

- Study the working principle of EV batteries.
- Analyze the major causes of battery degradation.
- Examine the relationship between battery degradation and driving range.
- Study the impact of battery health on user confidence and range anxiety.
- Propose a transparent battery health monitoring system for EV users.
- Recommend best practices to improve battery lifespan.

02 Overview of EV Batteries & Current Problem

Most EVs use Lithium-Ion batteries managed by a **Battery Management System (BMS)**, which continuously monitors the parameters below. The proposed transparency system extends these capabilities by surfacing meaningful battery health insights directly to users.

- | | |
|---------------------------------|-------------------------|
| ● State of Charge (SOC) | ● State of Health (SOH) |
| ● Battery Temperature | ● Cell Voltage Balance |
| ● Charging / Discharging Cycles | ● Battery Safety |

✗ Dashboards currently show

- Battery Percentage
- Estimated Remaining Range

✓ Users cannot easily access

- Battery Health (SOH)
- Remaining Battery Life
- Capacity Loss
- Battery Degradation History
- Fast Charging Impact
- Charging Pattern Analysis

This limited information makes it difficult for users to understand the true condition of their battery and increases uncertainty during long-distance travel.

03 Causes of Battery Degradation

Several operational and environmental factors accelerate the loss of usable battery capacity over time. The chart below ranks the primary contributors by their relative impact.

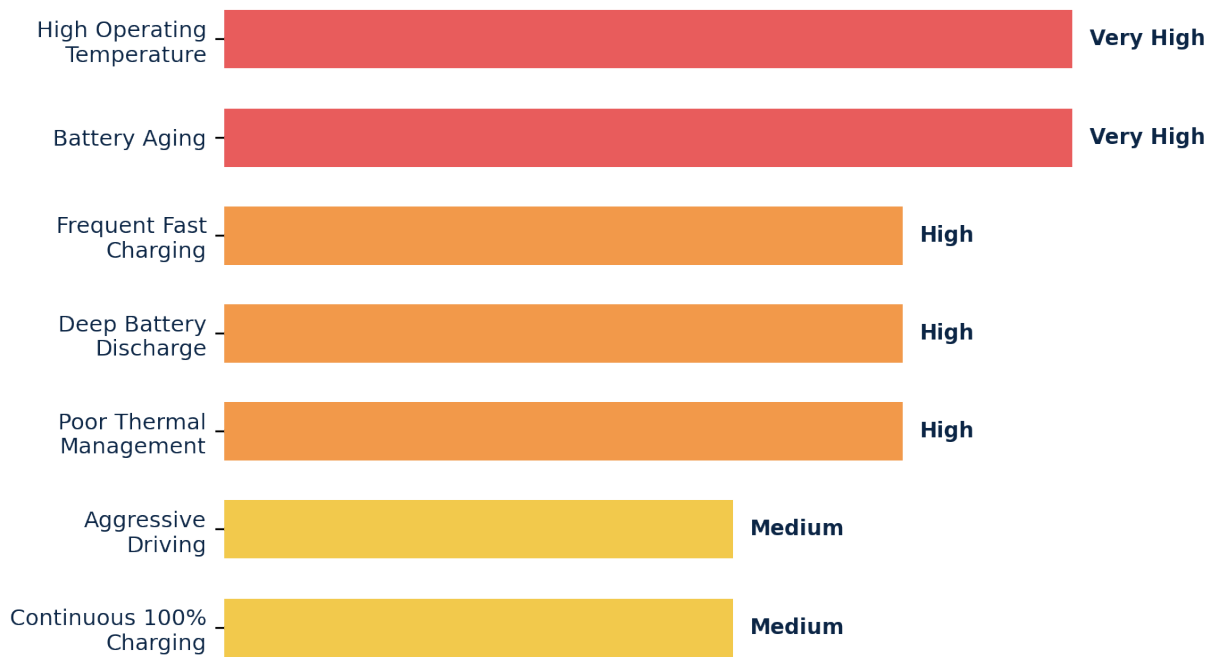


Figure 3.1 — Relative impact of key degradation factors

Factor	Impact on Battery
Frequent Fast Charging	High
High Operating Temperature	Very High
Deep Battery Discharge	High
Continuous 100% Charging	Medium
Battery Aging	Very High
Aggressive Driving	Medium
Poor Thermal Management	High

04 Battery Life Cycle Analysis

As charge cycles accumulate, State of Health (SOH) declines steadily, directly reducing the vehicle's usable driving range. The relationship is nearly linear across the observed cycle range.

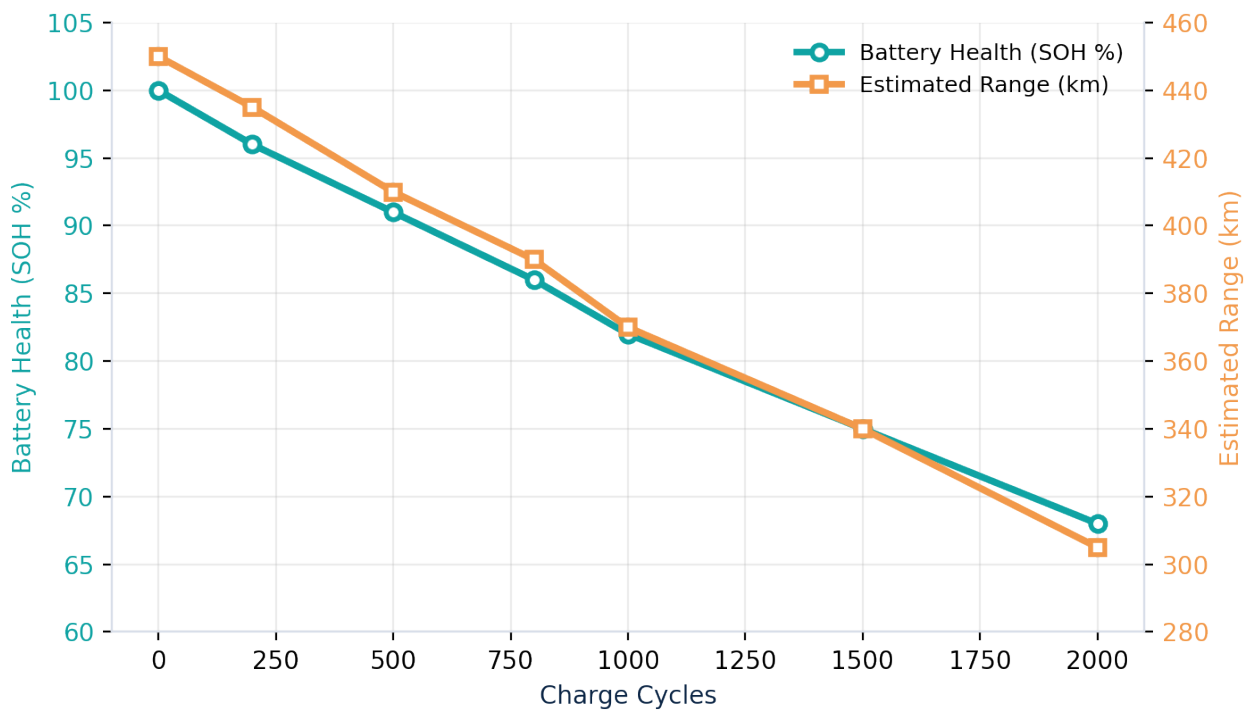


Figure 4.1 — Battery Health (SOH) and Estimated Range vs Charge Cycles

Charge Cycles	Battery Health (SOH)	Estimated Range (km)
0	100%	450
200	96%	435
500	91%	410
800	86%	390
1000	82%	370
1500	75%	340
2000	68%	305

Battery Health vs Driving Range & Performance

Battery Health	Available Range	Performance
100%	450 km	Excellent
90%	410 km	Very Good
80%	370 km	Good
70%	325 km	Moderate

Battery Health	Available Range	Performance
60%	280 km	Poor

05 Temperature & Charging Behaviour Effects

Operating temperature has one of the strongest effects on degradation rate. Charging habits — particularly avoiding extremes of state-of-charge — also play a major role in preserving battery life.

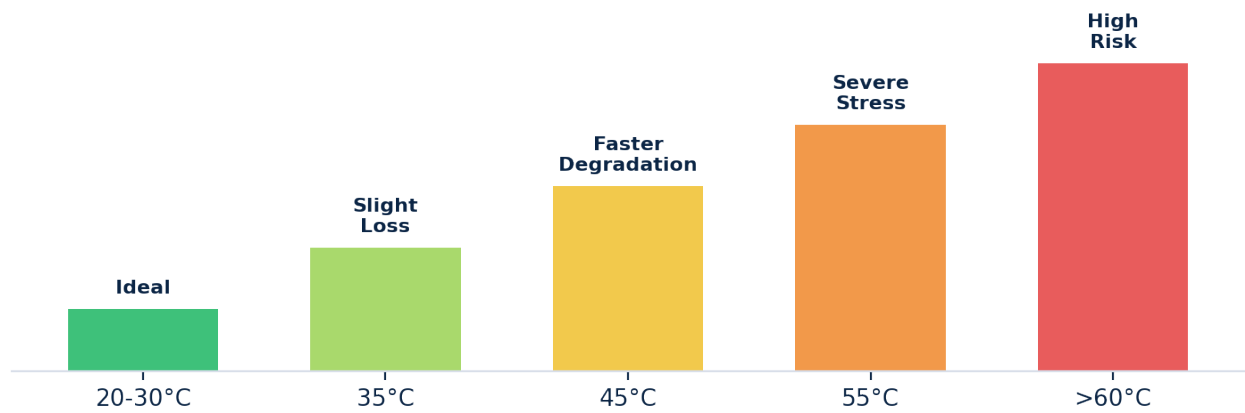


Figure 5.1 — Battery stress level by operating temperature

Battery Temperature	Effect
20–30°C	Ideal Operating Condition
35°C	Slight Capacity Loss
45°C	Faster Degradation
55°C	Severe Battery Stress
Above 60°C	High Risk of Battery Damage

Charging Behaviour Analysis

Charging Habit	Battery Life Impact
Charge between 20%–80%	Excellent
Daily Fast Charging	Poor
Overnight Charging to 100%	Moderate
Mixed Charging Pattern	Good

06 Range Anxiety Analysis

Range anxiety is the fear that an EV will run out of charge before reaching a charging station. This concern is often driven by **limited battery information** rather than the actual battery condition. A user survey (illustrative, for case study purposes) highlights the top concerns:

Percentage of Users Reporting Concern

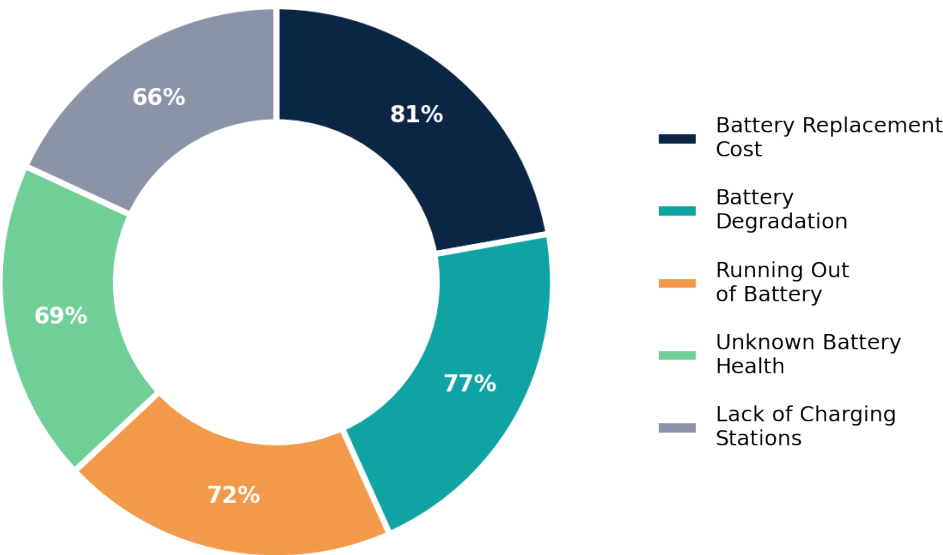


Figure 6.1 — Percentage of users citing each concern (illustrative survey data)

07 Proposed User Transparency System

The proposed dashboard provides detailed battery information instead of only showing battery percentage, giving users a complete and honest picture of battery condition.

■ Battery Health (SOH)	■ State of Charge (SOC)	■ Estimated Driving Range
■ Remaining Useful Battery Life	■ Battery Temperature	■ Charging Cycle Count
■ Fast Charging History	■ Battery Capacity Loss	■ Cell Health Status
■ Charging Recommendations	■ Battery Maintenance Alerts	

08 Comparison, Advantages & Limitations

Existing vs Proposed Dashboard

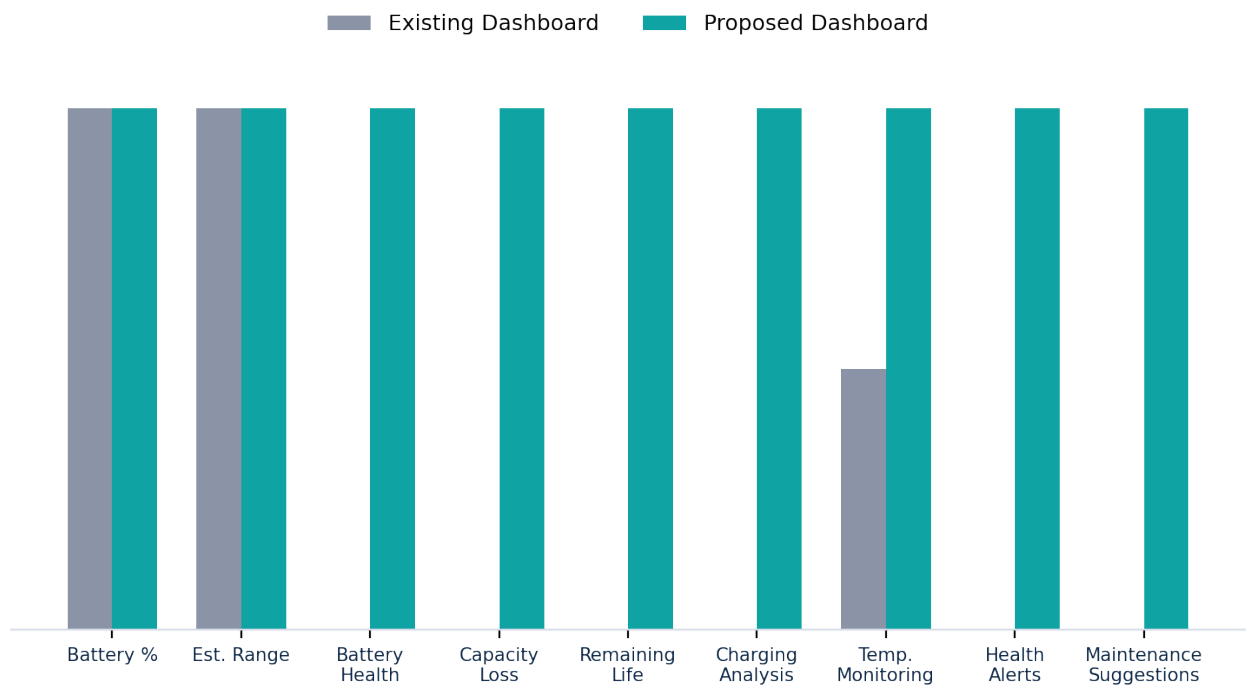


Figure 8.1 — Feature coverage: existing dashboard vs proposed system

Feature	Existing	Proposed
Battery Percentage	✓	✓
Estimated Range	✓	✓
Battery Health (SOH)	✗	✓
Capacity Loss	✗	✓
Remaining Battery Life	✗	✓
Charging Analysis	✗	✓
Temperature Monitoring	Limited	✓
Battery Health Alerts	✗	✓
Maintenance Suggestions	✗	✓

Advantages

- Reduces range anxiety with accurate battery information.
- Improves user confidence during long trips.
- Encourages healthy charging habits.
- Helps extend battery lifespan.
- Enables predictive maintenance.
- Improves resale value through battery health records.

Limitations

- Requires accurate battery sensor data.
- Degradation varies with climate and driving habits.
- AI-based predictions require continuous calibration.
- Additional software development increases cost.

09 Results & Conclusion

Key Results

- Battery degradation directly affects driving range and charging efficiency.
- High temperatures and frequent fast charging accelerate battery aging.
- Displaying SOH, remaining life, and charging insights helps users better understand battery condition.
- A transparent battery dashboard can reduce range anxiety, improve charging behavior, and increase confidence in EV ownership.

Conclusion

This case study highlights that battery degradation is a natural process that significantly influences EV performance, driving range, and user satisfaction. While current EV dashboards provide only basic information such as battery percentage and estimated range, they lack detailed insights into battery health and long-term performance.

The proposed **User Transparency System** addresses this gap by displaying Battery Health (SOH), remaining useful life, charging history, temperature effects, and maintenance recommendations. Such a system empowers users to make informed decisions, adopt better charging practices, reduce range anxiety, and maximize battery lifespan — ultimately improving the overall EV ownership experience.

Bottom line: Transparency in battery health data converts uncertainty into confidence — the single largest lever available to reduce range anxiety without changing the battery itself.